

An Evaluation of Text-Only and Text-3D Model Instruction Formats in Augmented Reality Training

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ABSTRACT

Augmented reality (AR) exhibits potential for delivering training instructions. Trainee learning and transfer can be influenced by the different extraneous cognitive loads implicit in text and graphical instruction formats. Our research evaluated this influence using a GoCube™ training procedure delivered in AR. Learning, transfer and user experience were evaluated using mental rotations, physiological ratings, eye gaze, facial expressions, memory recall and questionnaire responses. Results showed that trainees using text-only instruction were significantly faster in training and recall than their counterparts using a combined text and 3D-model instruction format. This correlated to mental rotation baselines, gaze shifts and cognitive load.

Index terms: Augmented reality, training instruction, gender.

1 INTRODUCTION

Augmented reality (AR) is increasingly used across a wide range of domains as a training platform due to numerous benefits such as instruction verification, trainee pacing and the ability to render a single animated analogous model of the workpiece in multiple configurations throughout training [1]. Learning and transfer are the pedagogical goals of the trainee. Learning is the acquisition of an ability to perform a procedure as trained. Transfer involves generalising that knowledge to similar procedures in the same domain.

Instructions can be presented in procedural (e.g. text) and example (e.g. graphics) formats [2]. Example dependency can result in learning and transfer deficiencies. Contrarily, the cognitive effort required to understand and carry out procedural instructions may benefit long term learning and transfer. In addition to this, gender-based disparities in mental rotation abilities amongst AR trainees have been reported [1], coinciding with calls for research focusing on larger gender balanced samples to control for bias [3]. Our work answers this call using a gender-balanced sample of sixty trainees to test a hypothesis that text and 3D model instruction formats can influence AR training. This was achieved using an AR training application on the HoloLens 2™ (HL2) to answer the research question; what is the influence of text-only and text combined with 3D model instruction formats on AR training? The trainees used the AR application to learn a 14-step GoCube™-based (see Fig. 1) mental rotation procedure [4]. The trainees'

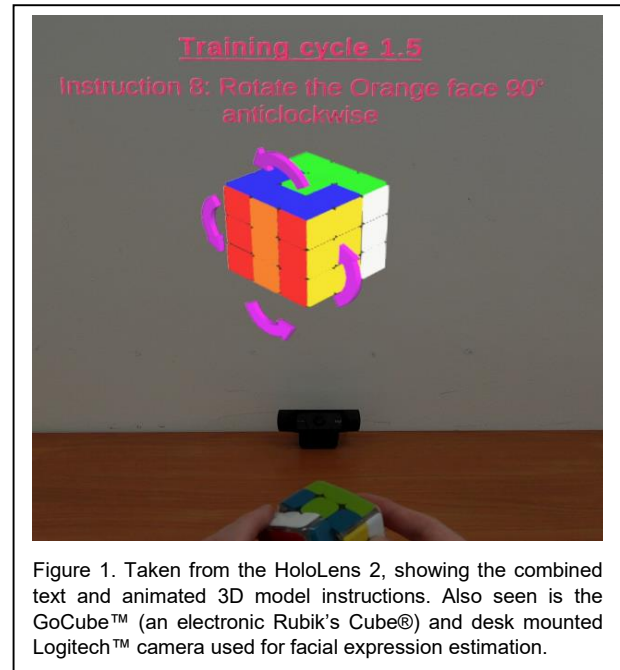


Figure 1. Taken from the HoloLens 2, showing the combined text and animated 3D model instructions. Also seen is the GoCube™ (an electronic Rubik's Cube®) and desk mounted Logitech™ camera used for facial expression estimation.

physiological ratings, facial expressions and eye gaze features were recorded in real-time during the evaluation to strengthen an understanding of the influence of instruction format on the trainees. The trainees completed post-training affect, NASA-TLX and modified usability questionnaires, which the interested reader can see in the supplemental materials.

Results showed that the text-only instruction format yielded faster instruction response times during training. A statistically significant correlation between slower training and slower recall from memory in the female trainees that used the 3D model means that the influence of instruction format on recall speed in the female trainees cannot be ruled out [1]. This was indicated by correlation to higher gaze shifts, heart rates and cognitive load during training in this cohort of trainees. Furthermore, positive transfer of AR GoCube™ training to general mental rotation abilities can also not be ruled out [4].

2 METHODOLOGY

In the independent samples study, the main test group (TG) had the benefit of an animated interactive 3D model instruction format as per Fig. 1. This was combined with text as recommended by [5]. The control group (CG) only had the use of the text instruction seen in Fig. 1. Cube-state was recorded on the HL2 during training and

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recall using one-way 2MBs low energy Bluetooth™. This facilitated the delivery of training and corrective instructions during training. This involved a repeating animation of 3D cube model face rotations accompanied by directional arrows in the TG environment. The text instructions consisted of cube face (white, yellow, orange, red, green and blue) and rotation direction (90° clockwise or anticlockwise) information as per Fig. 1.

The experimental protocol consisted of seven key phases. These were i) participant sampling, ii) screening, iii) instruction and headset calibration, iv) physiological baseline, v) practice, vi) task and vii) questionnaires. This research was approved by the Technological University of the Shannon, Midlands and Midwest Ethics Committee. The sample group had an age range from 19 to 62 years old with a mean age of 32. Each trainee was screened for visual acuity and colour perception. Phase vi (the task) consisted of training, waiting and recall sub phases. Training consisted of cycles comprising two halves that began and ended with the GoCube™ in the solved state. Training was self-paced, and the trainee could perform as many training cycles as needed to learn the procedure. Once the trainee indicated that they had learned the procedure, they had to replicate it from memory. This was done after a 30 second waiting period during which they performed arithmetic equations. This waiting period ensured recall from long term memory. The trainees' mental rotation abilities were recorded pre- and post-training using Vandenberg mental rotation tests. The trainees' heart rate, electrodermal activity and skin temperature were recorded from baseline to recall using the medical grade Empatica E4™ wrist-worn device. A desk-mounted Logitech™ video camera (see Fig. 1) was used in conjunction with OpenFace 2.0 to estimate facial expressions. Eye gaze dwell and shift rates were recorded on the HL2. Trainees completed the questionnaires (see supplemental material) to conclude the evaluation.

3 RESULTS

The difference in training instruction response times of 5s in the TG and 4s in the CG was statistically significant with $p = 0.05$. During recall from memory, there was a statistically significant difference between cube face rotation durations of 4 s for the females of the TG and 2s for females of the CG, with $p < 0.01$. Females of the TG had the slowest training instruction responses and the slowest cube face rotations during recall, as similarly reported in [1]. Both their training durations and recall rotations were accounted for by their mental rotation baseline (male: 8, female: 7) with $R^2 = 1.00$, $p < 0.01$. However, 46% of the variance in female TG recall rotations were also accounted for by their training instruction response times with $R^2 = 0.46$, $p < 0.01$. The direct correlation between recall and training durations precludes the dismissal of a direct relationship between training instruction format and speed of recall from memory in the female trainees of the TG. This was accompanied by the TG females reporting more cognitive effort invested during training than females in the CG (TG: 1.0, CG: -0.3, $p = 0.06$) on the modified usability questionnaire. This higher cognitive load accounted for 29% of the variance in higher gaze shift rates in TG females (male: 53, female: 66) with $R^2 = 0.29$, $p = 0.04$. Deviation from baseline of maximum heart rate in TG females was significantly higher than their CG counterparts during the practice phase (TG: 6 bpm, CG: -1 bpm, $p = 0.01$). This higher heart rate was indicated as higher stress by their significantly shorter inter-beat intervals (TG: 0.1s, CG: 0.6s, $p < 0.01$). Thirty-four percent of the variance in TG females maximum HR during training was accounted for by their quantity of training cycles (TG: 3, CG: 4, $p = 0.44$) with $R^2 = 0.34$, $p = 0.02$.

On the Self-Assessment Manikin component of the affect questionnaire (see supplemental materials), gender accounted for 35% of the variance in arousal (female: 1.3, male: 1.0, $p = 0.36$),

valence (female: 2.9, male: 2.4, $p = 0.11$) and dominance (female: 1.3, male: 1.6, $p = 0.56$), with $R^2 = 0.35$, $p < 0.01$.

A statistically significant difference of $p = 0.05$ in correct mental rotation abilities between CG male (9 correct) and female (6 correct) trainees at baseline was absent in the post-training Vandenberg test. A ceiling effect likely played a role in closing the gap. This does not preclude the possibility of positive transfer of AR GoCube™ training to general mental rotation abilities.

4 CONCLUSION

This work tested the hypothesis that text and 3D model instruction formats can influence AR training. A combined text and 3D model instruction format was compared against a text-only control in a GoCube™-based mental rotation training procedure delivered in AR. Results showed that the text-only instruction yielded faster instruction responses during training. This trend continued into recall, where the female trainees who used the 3D model during training were significantly slower performing cube face rotations from memory than their control group counterparts. Although the duration of their cube face rotations during training and recall correlated strongly to their lower correct mental rotation baselines, it also correlated moderately to their training instruction response times. This does not preclude a possible influence of the 3D model instruction format, not only on training duration, but also on speed of recall from memory in the female trainees. This result is strengthened by demonstrating how higher heart rate, gaze shift rate and cognitive load complemented an understanding of their higher stress and cognitive load levels. These surprising insights confirm the hypothesis that text and 3D model instruction formats can influence AR training by showing in this case that text-only instruction led to significantly faster training in general, and significantly faster recall from memory in female trainees. Additionally, this research has shown that AR mental rotation training may have played a role in closing a significant difference between the genders of the control group in general mental rotation baselines. Replication of this research using varying instruction formats and media may deepen an understanding of these results.

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